

The uptake of applied ecology

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Summary

1. We asked 229 authors who have published recently in the *Journal of Applied Ecology* (1999–2001) whether their papers made management or policy recommendations and whether they had evidence of consequent uptake.

2. A total of 108 respondents working in the UK (34%), Europe (30%), the Americas (12%), Australasia (11%), Asia (7%) and Africa (6%) reported on 110 papers. They represented agro-ecosystems (35%), temperate forests or woodlands (16%), savanna, grass or arid lands (11%), rivers or wetlands (10%), estuaries or marine systems (7%) and tropical forests (5%). The major organisms were invertebrates (27%), birds (24%), mammals (21%) and higher plants (21%). Topics apparently under-represented in recent coverage include ecosystem science, urban areas, soils, mountain systems, fish, amphibians and lower organisms such as algae.

3. Almost all papers (99%) carried recommendations and for 57% there was evidence of uptake in the broad categories of ‘environmental management or models’, ‘information, training and education’ and ‘monitoring and assessment’. Most uptake involved large geographical scales through habitat or species management plans (32% of cases), effects on reserve design or designation (6%), and effects on agri-environmental policy (5%). The development of further research (11%), the communication of methods to other ecologists (9%), the dissemination of recommendations to practitioners or agencies (7%), and uptake in training or education (5%) were important uses of information.

4. Prestige from publication in the *Journal of Applied Ecology* aided several authors in convincing end-users of research value. User involvement in research as participants or funders was widespread (> 42% of papers), a fact which almost certainly promotes uptake along with the parallel dissemination of management messages. We view applied issues as an important interface between end-users and ecologists of value to ‘both’ communities but suggest that improved communication will further benefit the sponsorship and application of ecological science.

5. The major reason offered for lack of uptake was that it was still too soon after publication (21% of respondents). Costs, difficulty of implementation, the scale of the problem, and ‘challenges to existing thinking’ each figured in more than one response.

6. For some respondents, papers were led by curiosity rather than the need for direct application. Several authors published in the *Journal* to share ideas internationally, or said that recommendations were general, conceptual or long-term rather than specific. The editors of the *Journal of Applied Ecology* recognize the seminal importance of contributions that affect policy incrementally and conceptually as much as those with specific application.

7. These data provide evidence that ecological science is aiding environmental management and policy across a wide range of regions, ecosystems and types of organisms; rather than merely detecting problems, applied ecology is offering solutions both directly and more diffusely through conceptual advance. We invite the user community to offer their own perspectives about the value of research-led publications such as this *Journal*, about how links between researchers and users might be strengthened, and about how the uptake of applied ecology might be further advanced.

Introduction

IMPACT OR UPTAKE?

On academic indicators of quality, the *Journal of Applied Ecology*, along with its sister journals from the British Ecological Society (BES), is fortunate in having sustained long-standing impact in the top tier of all ecological journals. It also offers one of the largest circulations and most international perspective among its competitors: in addition to being read widely on all continents, the Journal carries many contributions in which Europe, Australasia, the Americas, Asia and Africa figure prominently alongside those from the UK (see 'Results').

For ecology in general, however, and for the *Journal of Applied Ecology* as one of the leading outlets for applied research effort, academic indicators might reveal only part of our impact. In a world characterized by the need to derive environmental goods and services, and by the ecological problems that often result, ecologists must make contributions that are tangible and lasting. In other words, an important measure of the impact and value of ecological science is in the extent to which our research translates into utility. Part of the mission of applied journals like ours is not only to promote science excellence in ecology, but to take the subject into the wider and challenging domain of applied relevance.

Probably the clearest need for such ecological utility arises from the large number of organizations in all sectors – governmental, non-governmental and business – whose interests are focused on environmental resources. In predominantly rural areas such as Wales, a small country of 2.8 million people, recent estimates suggest that 17% of the total employment (= 169 000 full time equivalent jobs), 15% of good and services, and 9% of the nation's GDP of £26.7 billion sterling are dependent directly or indirectly on environmental use and management. Even in traditionally more industrial British regions, such as north-eastern England, around 6% of jobs are directly related to the environment; at over 51 000, this is at least as large as other major sectors such as construction, or the manufacture of motor vehicles, chemicals and IT equipment (The National Trust 2001). In both these cases, the majority of environmentally orientated employment is in the use of resources for tourism, agriculture, forestry, fishing, defence, water supply and extraction, but in Wales around 25% (c. 30 000) of the direct employment in this

sphere is in jobs that 'protect and enhance the environment'. Calculations of this sort are difficult. They are, however, just one minor illustration of how all of the global population of over 6 billion people ultimately depend on the use and production of environmental goods and services in all forms. Moreover, the growth of towns and cities worldwide means that ecological science has moved increasingly into urban as well as rural environments; for example, through the introduction of conservation biology into urban planning (Clergeau, Jokimäki & Savard 2001). The key point, nevertheless, is that these economic, planning and subsistence activities are at the heart of applied ecology. Along with a range of other professionals and partners, ecologists help to ensure that environmental use is sustainable; where it is not, they help to detect problems or, of even more value, they offer sound solutions that avoid environmental damage, restore losses and protect key features. Practitioners, user communities, decision-makers and policy-makers from grass roots to senior government administration benefit from our contributions in this respect.

For the *Journal of Applied Ecology*, one of the key questions is about how much the Journal's own papers contribute. This question is not only of academic interest because the answer has utility in its own right. It would allow feedback to those who sponsor or promote ecology about the wider impact of ecological research. It would inform our authors and readers about any additional benefits that arise collectively from the work we publish. It could offer examples and evidence to end-users of how the Journal might help their work. Finally, it would allow us to disseminate any good practice that improves the use and uptake of applied ecological research. Hitherto, however, no information of this type has been available. In July 2001, therefore, the *Journal of Applied Ecology* followed a recent example from *Conservation Biology* (Flashpohler, Bub & Kaplin 2000) and asked authors of recent papers about the use and uptake of their contributions into policy or management. Although very much a first step in what we see as a continuing debate, this editorial presents the results.

Methods

During July 2001, 229 corresponding authors who had published papers of all types in the *Journal of Applied Ecology* during the period 1999–2001 inclusive were contacted by e-mail. Authors were asked explicitly (i) whether they had made management or policy recommendations in their papers, and (ii) whether they

Table 1. Types of ecosystems and groups of organisms identified in the work of those responding. All values are percentages from the total number of times each system or group was represented

Ecosystem type	Percentage	Organism type	Percentage
Agroecosystems	35.5	Invertebrates	26.6
Temperate forests and woodlands	16.5	Birds	24.4
Savanna, grass and arid lands	10.7	Mammals	21.5
Rivers and other wetlands	9.9	Higher plants	20.7
Estuary and marine	6.6	Animals	2.9
Tropical forests	4.9	Fish	< 1
Uplands	4.1	Amphibians	< 1
Boreal/arctic	4.1	Ecosystems	< 1
All/any	3.3	Protozoa	< 1
Urban	1.6	Fungi	< 1
Soils	1.6		
Other	< 1		

Table 2. Papers from the *Journal of Applied Ecology* (1999–2001) in which authors said they made policy or management recommendations and offered evidence of use or uptake. Data available from *Conservation Biology* (1996–98) are shown for comparison

Percentage of papers with:	<i>Journal of Applied Ecology</i> (1999–01)	<i>Conservation Biology</i> (1996–98)
Management recommendations	99%	82%
Evidence of uptake or use	57% ¹	51%
<i>N</i> papers	110	?
<i>N</i> authors	108	145*
Author response rate	47% ²	30% ³

*Flashpohler *et al.* (2000) *Conservation Biology*, **14**, 1898–1902.

¹This value is 51% when ‘further research’ is removed as evidence of uptake (see text).

²Of 229 authors asked in total 1999–2001.

³Of 667 authors asked in total 1987–98.

had evidence of uptake into policy or management through their own work or anyone else’s. Authors were asked in open-ended questions what kind of use or uptake there had been and were asked to supply supporting evidence. They were asked for additional information on the kinds of organisms and ecosystems their work had involved, and they were invited to comment in any other way about the survey or about the *Journal of Applied Ecology*.

Results

THE SAMPLE

Responses were received from 108 individuals, or 47% of the sample contacted (cf. 29.7% in Flashpohler, Bub & Kaplin 2000), who in turn described 110 different papers. Respondents were 23% female and 77% male, and worked in the UK (34%), elsewhere in Europe (30%), Australasia (11%), North America (7%), Asia (7%), Africa (6%) and South America (5%). This geographical distribution closely follows the patterns of both submissions and publication by continent.

Over one-third of respondents were working on what they described as farmland or ‘agroecosystems’, but most of the world’s major ecosystems were also

represented (Table 1). There were also some unusual ecosystems, such as ‘fish processing systems’ (Wall, Howard & Bindu 2001), where ecological contributions are nonetheless economically important – in this case by assessing the risk of blowfly strike on fish during processing. Most work involved one or more taxonomic groups from the invertebrates, birds, mammals or higher plants (Table 1). Subjects that might be considered under-represented by virtue of their importance as management issues included urban, soil and mountain ecosystems. Also possibly under-represented was work on fishes, amphibians, lower organisms such as algae, and ecosystem science. Recent publications in the *Journal* have nevertheless covered these topics to some extent (e.g. Pakkala *et al.* 2001).

MANAGEMENT AND POLICY UPTAKE

Reflecting the *Journal*’s own guidelines, the vast majority of authors confirmed that their papers carried management or policy recommendations (see also Ormerod, Pienkowski & Watkinson 1999). More importantly, there was clear evidence of widespread use and uptake of ideas or methods among what are, essentially, a very recent batch of papers from the *Journal* (Table 2). Evidence of uptake existed for 57% of the

Table 3. The types of use or uptake in policy and management resulting from papers published in the *Journal of Applied Ecology* (1999–2001). The values are the total number of responding authors recording each category ($n = 81$)

Type of use	N of occurrence	Example papers
Environmental management or models		
Used in management plans or models	26	Palmer & Hester (2000), Bro <i>et al.</i> (2000), Stahl <i>et al.</i> (2001)
Guided the designation or design of reserves or forests	5	Sullivan <i>et al.</i> (1999), Whittingham <i>et al.</i> (2000)
Influenced agri-environment policy	4	Henderson <i>et al.</i> (2000), Milsom <i>et al.</i> (2000), Wolff <i>et al.</i> (2001)
Led to advice on pesticide use or subsidy	2	Van Den Berg & Soehardi (2000)
Used in policy review	2	Spurgeon <i>et al.</i> (1999)
Evaluated existing management procedures	2	Butterfield (1999), Oba <i>et al.</i> (2001)
Used by consultants in restoration schemes	2	Budelsky & Galatowitsch (2000)
Used in pesticide risk assessment	1	McKay <i>et al.</i> (1999)
Use in Biodiversity Action Plan	1	Grant <i>et al.</i> (1999)
Used to change legislation on plant provenance	1	Keller <i>et al.</i> (2000)
Used in development case-work	1	Gill <i>et al.</i> (2001)
Used by community groups in assessing biological control	1	Sheppard <i>et al.</i> (2001)
Information, training and education		
Generated further funding, research or trials	9	Thomas (2001), Shirley <i>et al.</i> (2001), Smith <i>et al.</i> (2001)
Methods used by other ecologists	7	Shea & Possingham (2000), Anciães & Marini (2000), Schneider (2001)
Translated into user-friendly brochures or other information for practitioners and agencies	6	Brøseth & Pedersen (2000), Manel <i>et al.</i> (2000), Di Guilio <i>et al.</i> (2001)
Used in training or educational materials	4	Cherrill & McClean (1999), Brøseth & Pedersen (2000)
Guided governmental research priorities	2	Siriwardena <i>et al.</i> (2000)
Example used in ministerial speech	1	Fensham & Holman (1999)
Used by specialist groups (e.g. IUCN)	1	Hoare (1999)
Monitoring and assessment		
Used in monitoring	3	Bilde <i>et al.</i> (2000), Hill <i>et al.</i> (2000), Davis <i>et al.</i> (2001)

sample overall and, even with the slightly questionable category of ‘use in further research’ disregarded, there was still evidence for 51% of authors. These values are remarkably similar to the 51% revealed by the recent survey of authors from *Conservation Biology* (Flashpohler, Bub & Kaplin 2000; Table 2). There were 81 individual instances of use or uptake from the 63 papers involved, several authors providing evidence of more than one type for some papers. Although these individual examples varied in character, three broad but interdependent categories were recognizable (Table 3).

First, papers that guided or evaluated ‘environmental management or models’ provided the single most widespread evidence of use. Most commonly, they involved management planning for habitats (Bühler & Schmid 2001), species of conservation importance (Fisher 2000; Suarez, Balbontin & Ferrer 2000), pest species (Brown & Singleton 1999), grazing systems (Fynn & O’Connor 2000; Humphrey & Patterson 2000; Palmer & Hester 2000), practices such as pesticide use (McKay *et al.* 1999), or features such as flood regimes in rivers (Robertson, Bacon & Heagney 2001). The next largest group in this category were contributions that had guided the designation or design of forests and reserves (Rushton *et al.* 2000). Next were those papers that had contributed at wider geographical scales by

influencing agri-environmental policy (Donald *et al.* 2001). In at least one instance, work published in the *Journal* had led to legal changes, in this case concerning plant-species provenance (Keller, Kollmann & Edwards 2000).

The second major recognizable category of uptake was in ‘information, training and education’, where the major use was in generating further funding or research (Table 3). Although on the surface not explicitly uptake for management, in some cases the research involved further trials necessary, for example, to control parasite vectors (Thomas 2001) or to develop applications (Shirley *et al.* 2001). Perhaps more tangibly, dissemination led to methods or information published in the *Journal of Applied Ecology* being used by other ecologists, including specialist policy groups such as at IUCN (Hoare 1999). Probably of even greater value in taking our work to those outside the subject, there were several instances where recommendations or key findings had been re-written in a form accessible to practitioners, users or environmental agencies. These sometimes involved translation where the user community or target audience did not have English as their first language. For example, Brøseth & Pedersen (2000) disseminated their findings through the Norwegian press and through leaflets for hunters and colleagues affected by their work. There will be

Table 4. Barriers to the uptake of applied ecology (this study) or conservation biology (Flashpohler, Bub & Kaplin 2000) volunteered by survey respondents to questionnaires

Barriers stated in this study	Barriers reported by Flashpohler <i>et al.</i> (2000)
Importance not recognized	Neglect of issue by agencies
Lack of institutional receptiveness	Insufficient training of managers/end users
Controversial nature of problem/public opposition	Public opposition
Time or economic costs of implementation*	Lack of funding
Political structures preventing uptake	Lack of infrastructure
Main purpose was curiosity-driven research*	
Difficulty of implementation*	
Challenged prevailing thinking*	
Awaiting results of other trails	
Need for better interface between ecologists and end-users*	

*Cited by more than one respondent.

many other cases outside English-speaking countries where the dissemination of management recommendations from the Journal requires this kind of translation, a feature raised by more than one respondent. One particularly important example of dissemination was an instance where information from a paper published in the Journal was used in a ministerial speech (Fensham & Holman 1999). Equally important in sustaining ecology into the future, there were clear examples where papers from the Journal had been used in training or education (Cherrill & McClean 1999). Throughout the world, many of the subscribers to the *Journal of Applied Ecology* are university or college libraries, so that education probably represents a substantially under-reported area of use.

The third and final category of uptake involved papers that had led to the use of ecology in monitoring or assessment, although these were perhaps surprisingly scarce among the data (Table 3). The few examples were nevertheless important; for example, the work by Hill *et al.* (2000) had made an important methodological contribution to the recent phase of the UK 'Countryside Survey'.

Further examples and categories of all types of use or uptake are recorded in Table 3.

FACTORS PREVENTING OR PROMOTING UPTAKE

Although not asked directly, several authors volunteered comments about factors that prevented or promoted uptake of their work. By far the majority among them (21% of all respondents) said that it was too soon after publication for there to be evidence of use. Others offered views about the difficulty of the issues involved; for example, with respect to the control of exotic and undesirable species or due to the sheer geographical scales at which applied ecologists must often work. The latter reveals a feature that is both a well recognized problem and an opportunity: the requirements of environmental management, and the systems or organisms that must be understood to advise it, often involve large-scale perspectives with all the attendant difficulties

(Ormerod & Watkinson 2000). At the same time, however, the clear evidence from this survey is that the majority of successes from applied ecology are at exactly the same large geographical scales at which environments are often managed: habitat or species planning; the designation or design of reserves and forests; agri-environmental management or policy.

Among our responses about factors preventing uptake were some interesting overlaps with the data from *Conservation Biology* (Table 4). Whether or not these were real problems, the perceived importance of issues, institutional receptiveness, challenges to prevailing thinking and public opposition in some respects are opportunities for dissemination and communication. Interestingly in this respect, several authors offered the view that peer-review, peer-recognition and international exposure in the *Journal of Applied Ecology* helped in convincing the user community of the quality and importance of their work. In other cases, appropriate translation for practitioners and dissemination was the key, since most end-users at whom recommendations are aimed are not regular Journal readers. Indeed, some authors were careful to point out that the policy effects of their papers in the Journal could not be separated from other means of disseminating the same information (e.g. Beukema & Cadee 1999).

One other feature likely to have promoted management uptake in this sample of papers were the institutions involved. Most often, these were universities, governmental and non-governmental research organizations. However, our own estimate from the responses was that at least 42% of our sample was either working in, related to, or sponsored by user organizations in government and non-governmental sectors that had some rôle in environmental management. Such links with key stakeholders are almost certainly important.

Perspectives

Exercises of this nature are notoriously difficult. They depend on a range of assumptions: for example, that those providing the primary information have made objective assessments about the effectiveness and use

of their work; or that we, as editors and analysts, have accurately quantified the information provided. Moreover, applied ecology takes many different forms. They range on the one hand from the very case- or site-specific investigations to those whose results or implications generalize more widely and over different spatio-temporal scales (Fleishman *et al.* 1999). Assessing use or uptake in these more general cases will be particularly difficult, a point well made by several respondents to our questionnaire. More than one researcher said that their recommendations were conceptual or long-term rather than specific. Others volunteered the view that their contributions were aimed at international dissemination to the Journal's readership rather than being directly aimed at any specific user group (Kleijn & Verbeek 2000; Siriwardena *et al.* 2000; Schneider 2001). The editors of the *Journal of Applied Ecology* fully accept the seminal value of contributions of this type, and we believe that impacts on policy or management can arise through diffuse, indirect and conceptual routes that change opinions incrementally as much as through case-specific applications. Our purpose is, after all, to blend academic and science excellence in ecology with management relevance. On the evidence of this survey, the Journal is effective in that it has sustained research leadership in ecology alongside what are now clear examples of uptake into management.

Ultimately, however, judgement about the practical effectiveness of research ecology requires that end-users, as well as research participants, find our outputs valuable. In this respect, we believe that the ecological community in general, and this Journal in particular, would benefit from the perspectives of those in the user community who depend on ecological information. Are the published outputs from Journals such as this as effective as we believe them to be? Can researchers improve their recommendations, or their methods of dissemination, to ensure fuller use of ecological science? Are we addressing the most appropriate questions?

To this end, the *Journal of Applied Ecology* is now taking two specific actions. First, we invite forum articles, particularly from end users, that broaden the debate about the subject of this editorial. Secondly, we are seeking and, in some much-needed cases, commissioning reviews and mini-reviews that explore more fully key subjects in the application of ecology. We invite informal approaches from contributors who will synthesize existing fields in ways that will be accessible to end users and researchers alike, or who will point to new and developing fronts of potential applied relevance. In these two ways and alongside other initiatives, we intend to continue to augment the quality of the *Journal of Applied Ecology* by increasing its appeal to researchers and end-users alike.

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